

Zehnora: where the model runs

Qwen3.6-35B-A3B on the university GPU PC. Every number in this document is measured on the PC or read from the model file. Measured on 23 September 2026.

34.66 B

parameters in the model (3 B used per token)

12,168 MiB

of model weights on the GPU (57 %)

9,146 MiB

of model weights in RAM, run by the CPU (43 %)

55.4 tok/s

measured writing speed (833 tok/s reading)

1. The machine

Component	Exact value	Role for the model
GPU	NVIDIA GeForce RTX 4070 Ti SUPER, 16,376 MiB VRAM, 285 W power limit, driver 591.86	Runs every layer's attention, the shared expert, 54 % of the expert weights, the output layer, and holds the conversation memory (KV cache)
CPU	AMD Ryzen 7 9700X, 8 cores / 16 threads; llama.cpp uses 8 threads	Runs the 46 % of expert weights that do not fit on the GPU, plus the token lookup
RAM	64 GB in the PC; 30,886 MiB (30.2 GiB) given to WSL/Docker	Holds the CPU's share of the model (9,146 MiB) and the file cache
SSD	1,007 GB disk, 933 GB free; model file 22,360,456,160 bytes	Stores the model file. It is read into memory once at start-up (about 25 s); the SSD does no work while the model answers

Software: llama.cpp server-cuda v0.4.1 in Docker (Windows 11 + WSL2 + Docker Desktop), automatic placement with `--fit on`, context window 65,536 tokens.

2. The model in one paragraph

Qwen3.6-35B-A3B is a **Mixture-of-Experts (MoE)** model. Each of its **40 layers** contains an attention part and **256 small "expert" networks**. For every token, a router picks **8 of the 256 experts** plus **1 shared expert**, so only about **3 billion** of the 34.66 billion parameters do work per token. That is why a model this size can run fast on one consumer GPU: the experts that are not needed for a token are never computed.

Property	Value	Property	Value
Total parameters	34,660,610,688	Layers	40
Experts per layer / used per token	256 / 8 + 1 shared	Attention type	10 full + 30 linear (Gated DeltaNet)
Hidden size	2,048	Attention heads (query / key-value)	16 / 2, size 256
Context the model supports	262,144 tokens	Context we run	65,536 tokens
File size	20.81 GiB	Average precision	5.158 bits per weight

Source: the GGUF file header (unsloth/Qwen3.6-35B-A3B-GGUF, file UD-Q4_K_XL, revision a483e9e), confirmed by llama.cpp's load log ("model params = 34.66 B", "file size = 20.81 GiB (5.16 BPW)").

3. Where every part of the model lives

At start-up llama.cpp measures free GPU memory and places as much of the model on the GPU as fits, keeping about 1 GiB free as a safety margin. Its own report: *"41 layers (20 overflowing), 13,860 MiB used, 1,160 MiB free"*. All 40 layers and the output layer are on the GPU. In 20 of those layers the experts do not fully fit, so part of their expert weights stays in RAM and is computed by the CPU.

GPU (VRAM) CPU (RAM)

Model weights: 21,314 MiB in total



Routed experts only: 18,760 MiB



All other parts (attention, shared expert, output head, norms: 2,038 MiB) are 100 % on the GPU; only the 515 MiB token table is on the CPU.

Part of the model	What it does	GPU	CPU / RAM	Precision
Token embedding table	Turns each token id into numbers (a table lookup)	0	515 MiB	8.5-bit (Q8_0)
Attention (10 full + 30 linear layers)	Lets each token look at the conversation so far	all	0	8.5-bit (Q8_0)
Router	Chooses the 8 experts for each token	all	0	32-bit (F32)
Shared expert	Runs for every token, next to the 8 chosen experts	all	0	8.5-bit (Q8_0)
Routed experts (256 per layer)	The bulk of the model's knowledge; 8 used per token per layer	10,130 MiB	8,631 MiB	Q4_K 65 %, Q5_K 32 %, Q6_K 3 %
Output head	Turns the final numbers into the next token's probabilities	all	0	8.5-bit (Q8_0)
Dense parts together	Attention + shared expert + output head + norms	2,038 MiB	0	
All model weights		12,167.74 MiB	9,146 MiB	5.158 bits avg

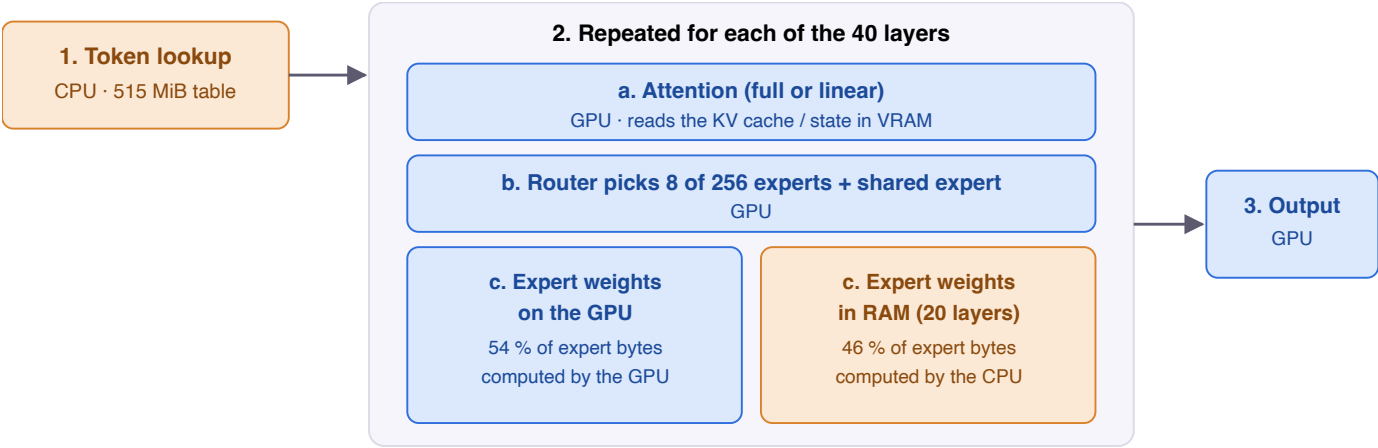
3b. Everything else in memory while the model runs

Item	Purpose	GPU	CPU / RAM
KV cache (F16, 65,536 tokens, 10 full-attention layers)	The model's memory of the current conversation	1,280.00 MiB	0
Linear-attention state (Gated DeltaNet, 40 layers)	Running summary used by the 30 linear-attention layers	62.81 MiB	0
Compute buffers	Scratch space for the calculations	350.03 MiB	72.02 MiB
llama.cpp total		13,860.6 MiB	9,218 MiB
CUDA driver context	NVIDIA runtime (difference to nvidia-smi)	273 MiB	
GPU memory in use (nvidia-smi)		14,134 of 16,376 MiB	

Source: llama.cpp load log at debug level (model-placement.sh): "CUDA0 model buffer size = 12167.74 MiB", "CUDA0 KV buffer size = 1280.00 MiB", "CUDA0 RS buffer size = 62.81 MiB", "CUDA0 compute buffer size = 350.03 MiB", "CUDA_Host compute buffer size = 72.02 MiB", fit result "Host 9,218 = 9,146 model + 72 compute". The expert split is calculated from these: 12,167.74 – 2,038 (dense on GPU) = 10,130 MiB; 9,146 – 515 (token table) = 8,631 MiB.

4. What happens for one token

Writing an answer means producing one token at a time. For every token, the work goes through the same path:



The GPU does the attention, routing, shared expert, output and its share of the experts. In the 20 layers where part of the experts live in RAM, the CPU computes those experts and hands the result back to the GPU. The SSD is not involved after start-up: the whole model is already in VRAM + RAM.

5. Measured speed and load

Test (llama.cpp's own timings)	Speed	GPU busy (avg / max)	GPU power (avg / max)	CPU (model)
Write 512 tokens	55.4 tokens/s	30.4 % / 41 %	56.7 W / 98.9 W	706 % ($\approx$ 7 of 8 threads)
Read a 9,800-token prompt	833.4 tokens/s	51.5 % / 72 %	86.6 W / 115.8 W	123 %
Real chats from the portal (server log)	52–61 tokens/s			
Coding answer with thinking (3,013 tokens)	52.24 tokens/s			
Idle (model loaded, no request)	–	0 %	4.5 W	0.04 %

What limits the speed. While writing, the CPU is almost fully busy (about 7 of its 8 threads) computing the experts that live in RAM, while the GPU is only about 30 % busy and waits for them. The writing speed is therefore set by the CPU and RAM part, not by the GPU. Reading a long prompt is different: many tokens are processed together, so the GPU does most of the work (833 tokens/s).

6. Why this setup

- **Quality:** the model in its original 16-bit form needs about 69 GB ($34.66 \text{ B} \times 2$ bytes). In this 5.16-bit version it needs 21 GB, which fits in the GPU's 16 GB plus the PC's RAM. A smaller model that fits fully on the GPU would be faster but is generally weaker at coding.
- **Speed:** because only 3 B parameters work per token, moving 46 % of the experts to RAM still gives 55 tokens/s, faster than people read.
- **Headroom:** 1,160 MiB of VRAM is kept free on purpose so the GPU never runs out of memory during long conversations.

Benchmarks: `zehnora/scripts/server/model-report.sh`, GPU and CPU sampled while each test ran. Full records: `zehnora/docs/MODEL-EVALUATION.md`.